The two-row $d = 4$ law for all $b \equiv 0 \pmod 4$: a 2-adic valuation identity

Clio

2026-06-08

Abstract

We close the case $b \equiv 0 \pmod 4$ of the two-row $d = 4$ fiber-vanishing law $G_{(2m-b,b)}(i) = 0 \iff (2,2)$. Together with the previously proved class $b \equiv 1 \pmod 4$, this establishes the law unconditionally for the full residue classes $b \equiv 0, 1 \pmod 4$ — half of all b. The proof is the exact 2-adic identity $v_2(I_b(m)) = v_2((m)_{R+1}) - v_2(R!)$, valid for every integer m. In contrast with the $b \equiv 1$ case — where a single term of the binomial expansion strictly dominates — here, for odd m, exactly three terms attain the minimal valuation; the law survives because *three odd numbers sum to an odd number*. The parity of the number of valuation-minimal terms is the operative invariant.

1 Setup (proved in prior cycles; not re-derived)

Fix $\lambda = (2m - b, b)$ with $0 \le b \le m$, put $s = 1 + i$, $P(u) = (1 + su + u^2)^m$, and

$$I_b(m) := \operatorname{Im} G_{(2m-b,b)} = \operatorname{Im}\,[u^b]\big((1-u)P(u)\big) \in \mathbb{Z}.$$

Earlier work establishes the following, which we take as given.

- **Exact expansion (E).** Writing $1 + su + u^2 = (1 + u^2) + su$ and extracting imaginary parts,

$$I_b(m) = \sum_{j=1}^{b} \tau_j(m), \qquad \tau_j(m) = \binom{m}{j} \operatorname{Im}\big((1+i)^j\big)\,\big(T(b-j) - T(b-1-j)\big),$$

  where $T(\ell) = \binom{m-j}{\ell/2}$ for ℓ even and $T(\ell) = 0$ for ℓ odd. Of the two extractions exactly one is nonzero (their indices $b-j$, $b-1-j$ are consecutive), so $T(b-j) - T(b-1-j) = \pm\binom{m-j}{\beta_j}$ with $\beta_j := \lfloor (b-j)/2 \rfloor$.

- **The complex factor.** $\operatorname{Im}\big((1+i)^j\big) = 2^{j/2}\sin(j\pi/4) \in \mathbb{Z}$, with

$$v_2\Big(\operatorname{Im}\big((1+i)^j\big)\Big) = \left\lfloor \tfrac{j}{2} \right\rfloor \quad (j \not\equiv 0 \bmod 4), \qquad \operatorname{Im}\big((1+i)^j\big) = 0 \quad (j \equiv 0 \bmod 4).$$

  In particular $\operatorname{Im}((1+i)^j)$ is odd iff $j = 1$.

- **Forced roots and the reduction.** $I_b(m)$ is an integer-valued polynomial in m whose only integer roots are the forced ones $m \in \{0, 1, \dots, R\}$, $R := \lfloor (b-1)/2 \rfloor$. Writing $I_b(m) = (m)_{R+1}\,Q_b(m)$ with $(m)_{R+1} = \prod_{r=0}^{R}(m - r)$, the two-row law is exactly

$$(\star) \qquad Q_b(m) \neq 0 \;\text{ for every integer } m \ge b.$$

  Since $m \ge b > R$ makes $(m)_{R+1} \neq 0$, $(\star)$ is equivalent to $I_b(m) \neq 0$ for $m \ge b$.

Throughout this note $b \equiv 0 \pmod 4$, say $b = 4t$. Then

$$R = \frac{b-2}{2} = 2t - 1 \text{ is } odd, \qquad R + 1 = \frac{b}{2} \text{ is even.}$$

The oddness of R is the structural fact that drives everything below.

Main result.

Theorem 1. *Let $b \equiv 0 \pmod 4$ and $R = (b-2)/2$. For every integer m,*

$$\text{(V)} \qquad v_2\big(I_b(m)\big) = v_2\big((m)_{R+1}\big) - v_2(R!).$$

Consequently $v_2(Q_b(m)) = -v_2(R!)$ is constant, so $Q_b(m) \neq 0$ for every integer m; in particular $(\star)$ holds and the two-row $d = 4$ law holds for all $b \equiv 0 \pmod 4$.

The same identity (V) was proved last cycle for $b \equiv 1 \pmod 4$, so **the two-row $d = 4$ law now holds for all $b \equiv 0, 1 \pmod 4$.**

2 The $j = 1$ term and the abbreviation a

For b even, $b - 1$ is odd and $b - 2$ is even, so in τ_1 the surviving extraction is $T(b-2) = \binom{m-1}{R}$ (here $\beta_1 = \lfloor (b-1)/2 \rfloor = R$), giving

$$\tau_1(m) = -\, m\binom{m-1}{R} = -(R+1)\binom{m}{R+1} = -\frac{(m)_{R+1}}{R!}.$$

Hence, abbreviating

$$a := v_2\big(\tau_1(m)\big) = v_2\big((m)_{R+1}\big) - v_2(R!),$$

the right-hand side of (V) is exactly a. The theorem is the statement $v_2(I_b(m)) = a$.

3 Term valuations: an exact decomposition

Fix j with $\tau_j \neq 0$, i.e. $j \not\equiv 0 \pmod 4$, and put $h := \lfloor j/2 \rfloor$. We compute $v_2(\tau_j) - a$ exactly.

Since b is even, $\beta_j = \lfloor (b-j)/2 \rfloor$ and one checks $s_j := j + \beta_j = R + 1 + h$ in both parities of j (for $j = 2h$: $\beta_j = R + 1 - h$; for $j = 2h+1$: $\beta_j = R - h$). The Vandermonde identity $\binom{m}{j}\binom{m-j}{\beta_j} = \binom{m}{s_j}\binom{s_j}{j}$ gives

$$v_2(\tau_j) = v_2\binom{m}{s_j} + v_2\binom{s_j}{j} + \left\lfloor \tfrac{j}{2} \right\rfloor.$$

Telescoping $\dfrac{\binom{m}{s_j}}{\binom{m}{R+1}} = \prod_{t=1}^{h} \dfrac{m - R - t}{R + 1 + t}$ (note $s_j - (R+1) = h$) and subtracting $a = v_2(R+1) + v_2\binom{m}{R+1}$ yields, after cancelling the common $v_2\binom{m}{R+1}$,

$$\boxed{v_2(\tau_j) - a \;=\; D_0(j) + S_j(m), \qquad S_j(m) := \sum_{t=1}^{h} v_2(m - R - t),}$$

where the m-independent offset is

$$D_0(j) = h + v_2\binom{s_j}{j} - \sum_{t=1}^{h} v_2(R+1+t) - v_2(R+1) = h + v_2\binom{s_j}{j} - \big(v_2((R+1+h)!) - v_2(R!)\big).$$

Using $s_j = R+1+h$, so $\binom{s_j}{j} = \dfrac{(R+1+h)!}{j!\,\beta_j!}$, the factor $(R+1+h)!$ cancels and $D_0(j) = h + v_2(R!) - v_2(j!) - v_2(\beta_j!)$. Evaluating the two parities with $v_2((2h)!) = v_2((2h+1)!) = h + v_2(h!)$:

Lemma 1. *For $\tau_j \neq 0$ and $h = \lfloor j/2 \rfloor$,*

$$D_0(j) = \begin{cases} v_2\dbinom{R}{h}, & j \text{ odd,} \\ v_2\dbinom{R}{h-1} - v_2(h), & j \text{ even.} \end{cases}$$

Proof. *$j = 2h+1$ odd:* $\beta_j = R - h$, so $D_0 = h + v_2(R!) - \big(h + v_2(h!)\big) - v_2((R-h)!) = v_2(R!) - v_2(h!) - v_2((R-h)!) = v_2\binom{R}{h}$.

$j = 2h$ even: $\beta_j = R + 1 - h$, so $D_0 = h + v_2(R!) - \big(h + v_2(h!)\big) - v_2((R+1-h)!) = v_2(R!) - v_2(h!) - v_2((R+1-h)!)$. Since $(R+1-h)! = (R-h)!\,(R+1-h)$ we get $D_0 = v_2\binom{R}{h} - v_2(R+1-h)$. Finally the elementary identity $h\binom{R}{h} = (R+1-h)\binom{R}{h-1}$ — both sides equal $\frac{R!}{(h-1)!\,(R-h)!}$ — gives, on taking v_2, $v_2\binom{R}{h} - v_2(R+1-h) = v_2\binom{R}{h-1} - v_2(h)$. ☐

4 No term dips below τ_1

Lemma 2 ((P1))**.** *For every j with $\tau_j \neq 0$ and every integer m, $v_2(\tau_j) \geq a$, i.e. $D_0(j) + S_j(m) \geq 0$.*

Proof. For j odd, $D_0(j) = v_2\binom{R}{h} \geq 0$ and $S_j \geq 0$, done.

For j even, the h integers $m-R-1, m-R-2, \ldots, m-R-h$ are consecutive, so their product is divisible by $h!$; hence

$$S_j(m) = v_2\Big(\textstyle\prod_{t=1}^{h}(m-R-t)\Big) \;\geq\; v_2(h!) \;\geq\; v_2(h).$$

By Lemma **??**, $D_0(j) = v_2\binom{R}{h-1} - v_2(h) \geq -v_2(h)$. Adding, $D_0(j) + S_j \geq -v_2(h) + v_2(h) = 0$. ☐

5 Exactly which terms tie

Lemma 3 (tie classification)**.** *Among $j \geq 2$ with $\tau_j \neq 0$, the equality $v_2(\tau_j) = a$ holds for some m only if $j \in \{2,3\}$. Moreover, for R odd:*

$$v_2(\tau_2) = v_2(\tau_3) = a \iff m \text{ odd}, \qquad v_2(\tau_2), v_2(\tau_3) > a \iff m \text{ even},$$

and for every $j \geq 4$ with $\tau_j \neq 0$ one has $v_2(\tau_j) > a$ for all m.

Proof. By Lemma **??** a tie means $D_0(j) + S_j(m) = 0$. We bound S_j from below.

j even, $h \geq 3$: $S_j \geq v_2(h!) = v_2(h) + v_2((h-1)!) \geq v_2(h) + 1$, because $(h-1)!$ is even for $h \geq 3$. With $D_0(j) \geq -v_2(h)$ this gives $D_0(j) + S_j \geq 1 > 0$: strict.

j even, $h = 2$: then $j = 4 \equiv 0 \pmod 4$, so $\tau_4 = 0$ — excluded.

j odd, $h \geq 2$ (i.e. $j \geq 5$): the list $m-R-1, \ldots, m-R-h$ contains two consecutive integers, one even, so $S_j \geq 1$; with $D_0(j) \geq 0$, $D_0(j) + S_j \geq 1 > 0$: strict.

Thus the only $j \geq 2$ that can tie are $j = 2$ ($h = 1$) and $j = 3$ ($h = 1$). For both, $S_j = v_2(m-R-1)$, and by Lemma **??**, using R odd,

$$D_0(2) = v_2\binom{R}{0} - v_2(1) = 0, \qquad D_0(3) = v_2\binom{R}{1} = v_2(R) = 0.$$

Hence $v_2(\tau_2) - a = v_2(\tau_3) - a = v_2(m-R-1)$. Since R is odd, $m-R-1$ is odd iff m is odd; therefore both differences vanish iff m is odd, and are ≥ 1 iff m is even. ☐

6 Proof of the Theorem

By Lemma **??**, $v_2(\tau_j) \geq a$ for all j, so $N(m) := I_b(m)/2^a = \sum_j \tau_j(m)/2^a$ is an integer, and modulo 2 only the *tying* terms (those with $v_2(\tau_j) = a$) survive:

$$N(m) \equiv \sum_{j\colon v_2(\tau_j)=a} \frac{\tau_j(m)}{2^a} \pmod 2.$$

Each summand $\tau_j/2^a$ is odd (its 2-adic valuation is exactly 0). By Lemma **??**:

- If m is even, the only tying term is $j = 1$, so $N(m) \equiv 1 \pmod 2$.
- If m is odd, the tying terms are exactly $j \in \{1, 2, 3\}$, so $N(m)$ is a sum of *three* odd integers, hence $N(m) \equiv 1 \pmod 2$.

In both cases $N(m)$ is odd, i.e. $v_2(I_b(m)) = a = v_2((m)_{R+1}) - v_2(R!)$, which is (V). For $m \geq b > R$ the right-hand side is finite, so $I_b(m) \neq 0$; equivalently $Q_b(m) \neq 0$. This is $(\star)$, the two-row $d = 4$ law for $b \equiv 0 \pmod 4$. □

Remark 1. The mechanism differs sharply from $b \equiv 1 \pmod 4$. There $R + 1$ is odd, every τ_j with $j \geq 2$ satisfies $v_2(\tau_j) > a$ strictly, and $I_b/2^a$ reduces to the single odd term $\tau_1/2^a$. For $b \equiv 0 \pmod 4$ the factor $R+1$ is even, which lowers the offsets so that, for odd m, the terms $j = 2, 3$ *tie* τ_1. The law is rescued not by domination but by counting: the number of valuation-minimal terms is odd (1 for even m, 3 for odd m), so their sum keeps an odd leading 2-adic digit. The oddness of R — a direct consequence of $4 \mid b$ — is what forces both $D_0(2) = D_0(3) = 0$ and pins the tie to the parity of m.

Remark 2. Equivalently, over $\mathbb{F}_2$ the monic irreducible part q_b of Q_b satisfies $q_b(m) \equiv (m^2 + m + 1)^{\lfloor (b-1)/4 \rfloor} \pmod 2$ for $b \equiv 0, 1 \pmod 4$ (verified $b \leq 40$); since $m^2 + m + 1$ is odd at every integer this re-expresses "no integer root." Theorem **??** proves the underlying non-vanishing directly via (V), without needing the polynomial factorization shape.

Verification

The decomposition of §**??**, the closed forms of Lemma **??**, the no-dip bound (P1), the tie classification, and the identity (V) were checked exactly for all $b \equiv 0 \pmod 4$ with $4 \leq b \leq 64$, over $m \in [b, b+20]$ together with 40 random $m \leq 10^6$ per b (24,480 term-level checks; `verify.py`: "ALL OK").